

Size and density of split matrix according to species

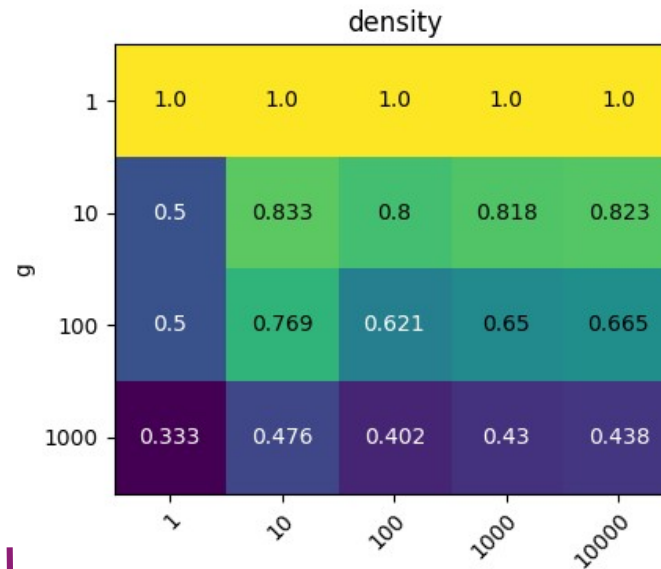
Marie Picard

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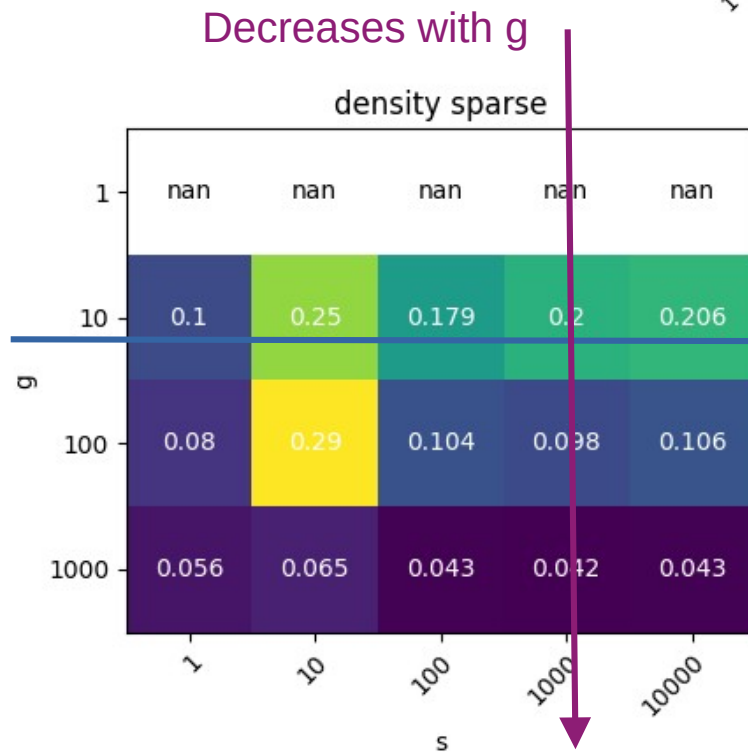
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February 13, 2026

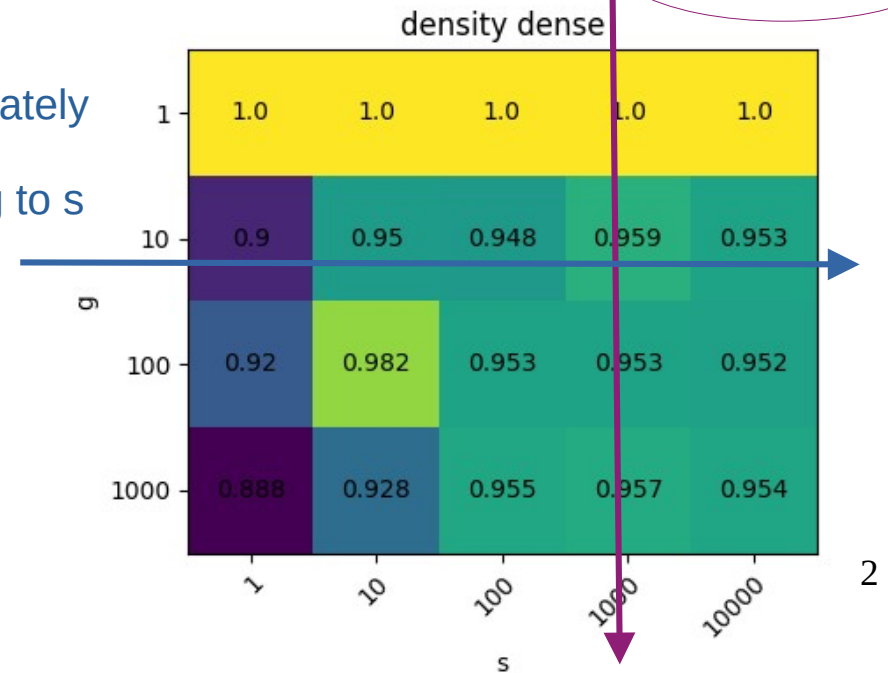
Quick recap of past results



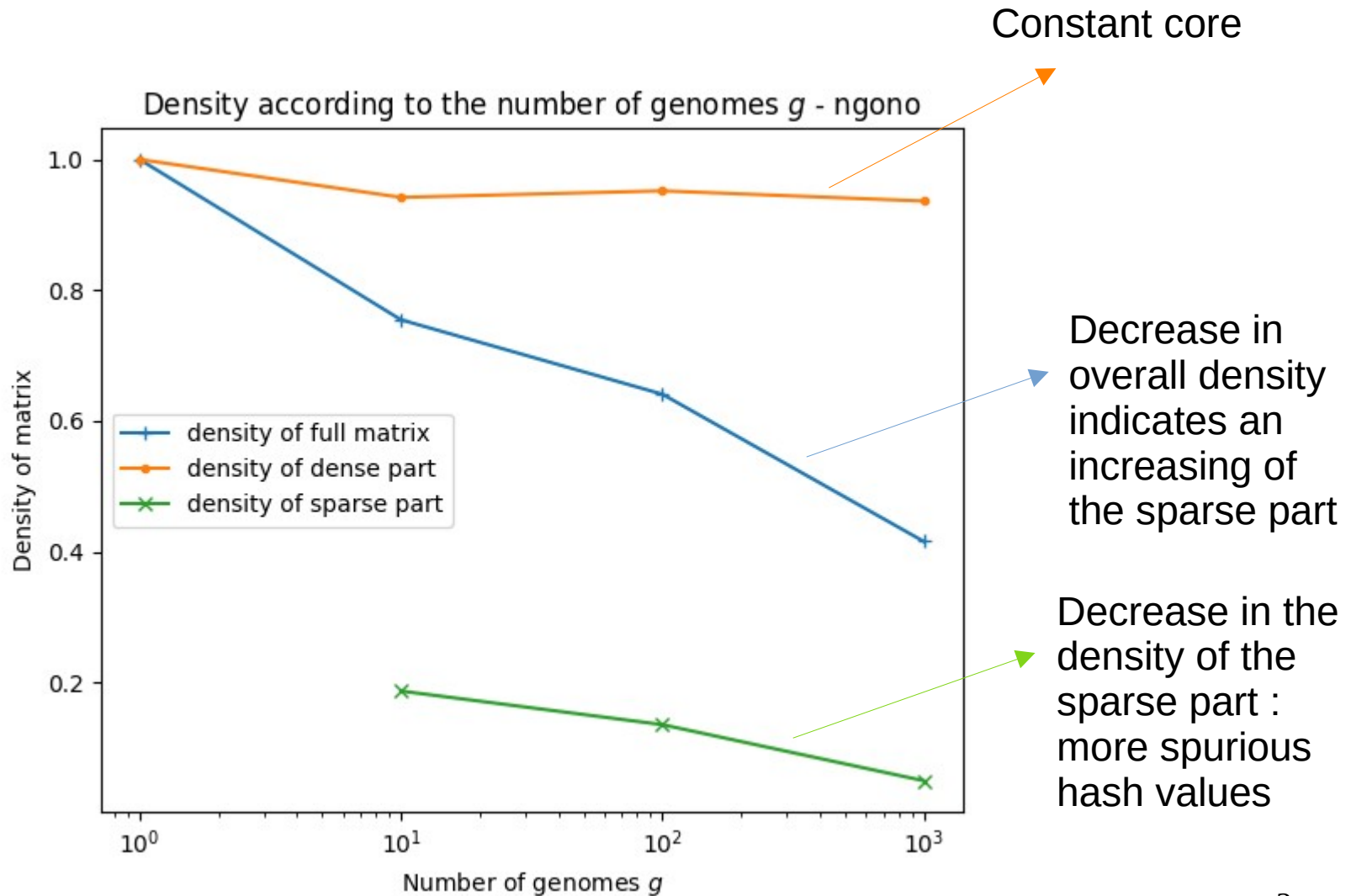
Conjecture : a “core” of hashes that corresponds to the dense part (whose number of rows doesn’t change much with g), and a part of spurious hashes



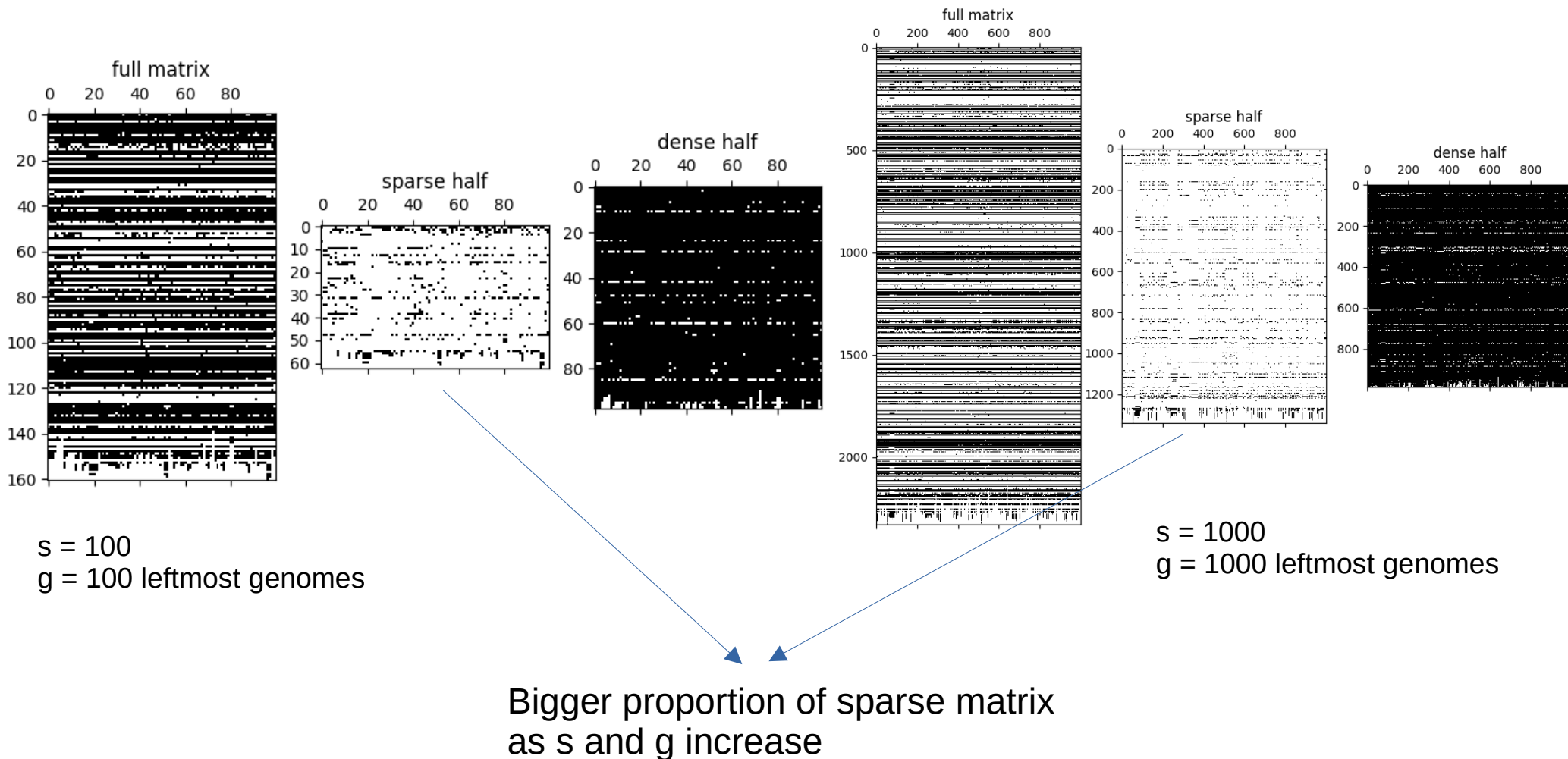
Approximately constant according to s



Quick recap of past results



Quick recap of past results



Questions :

- 1) How does the density change according to the species ?
- 2) How big is each part according to g and s ?

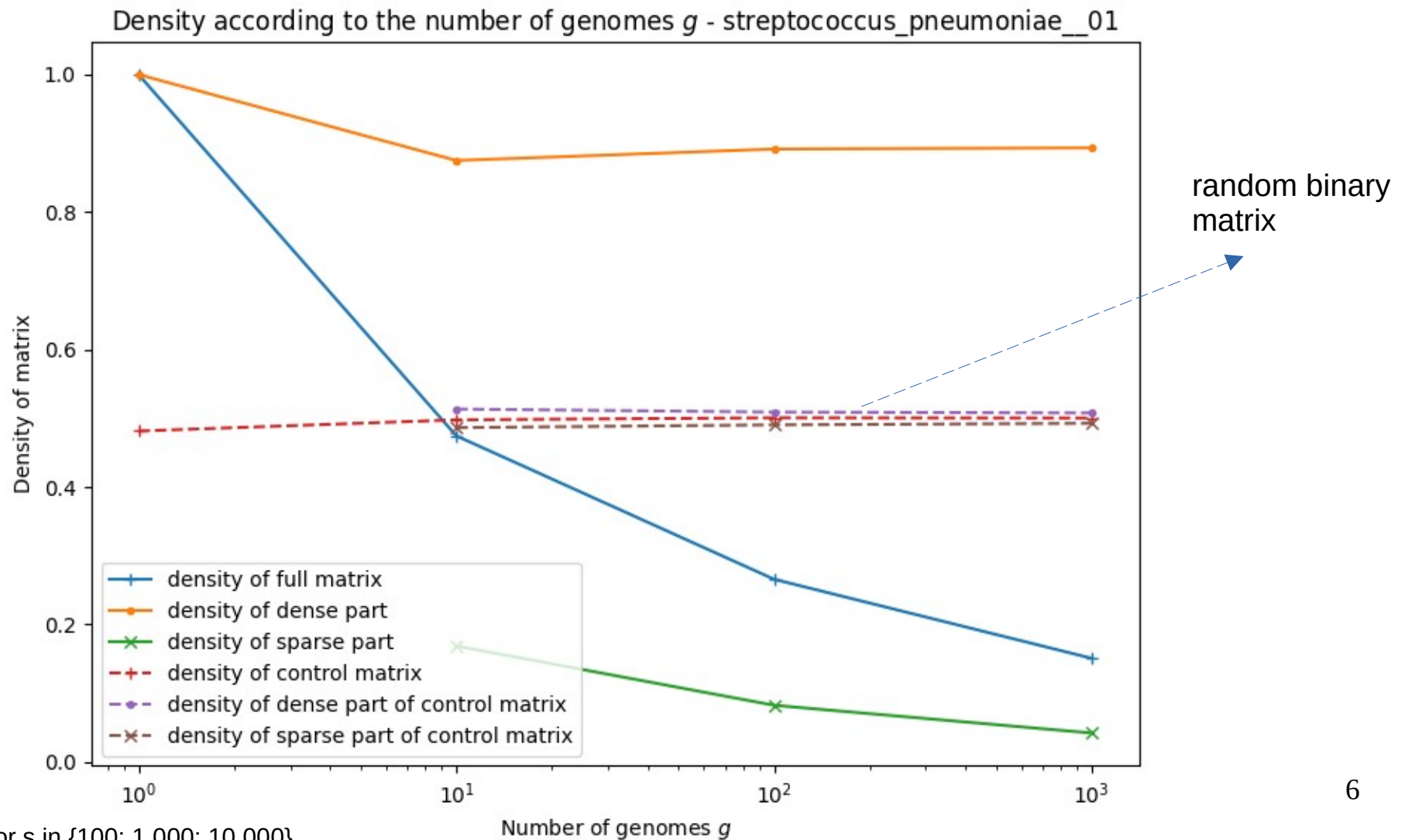


Number of rows

“core idea” hypothesis :

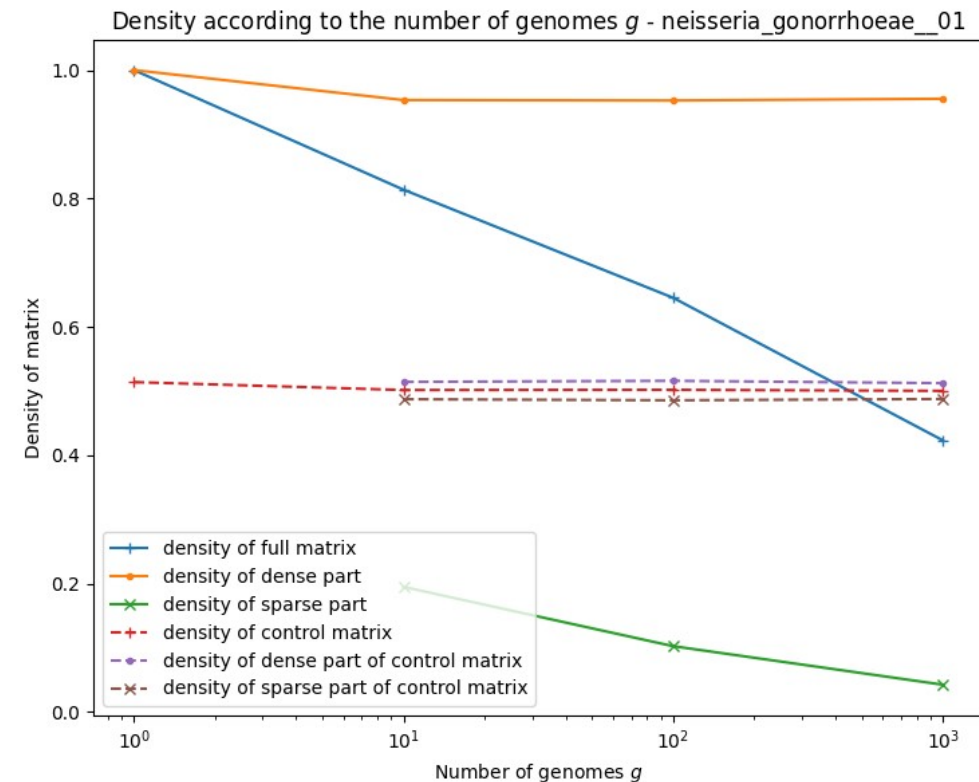
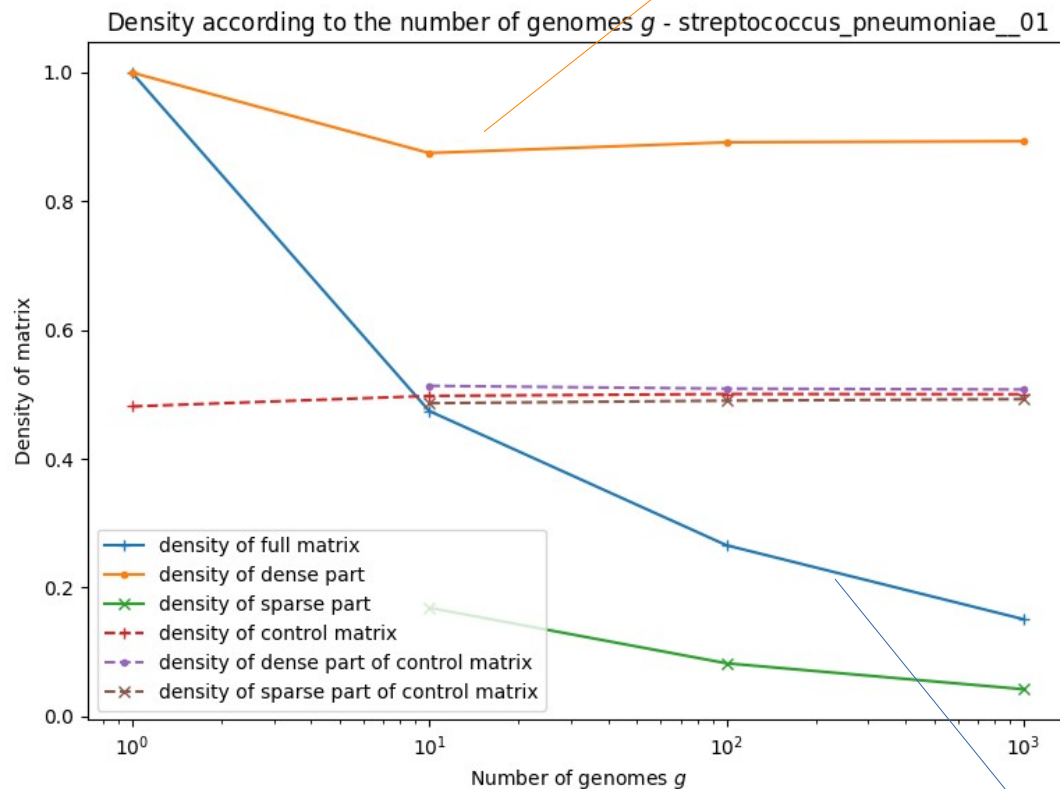
- The number of rows of the dense part does not change much with g
- The number of rows in the sparse part increases linearly in $(s \cdot g)$

Density of other species



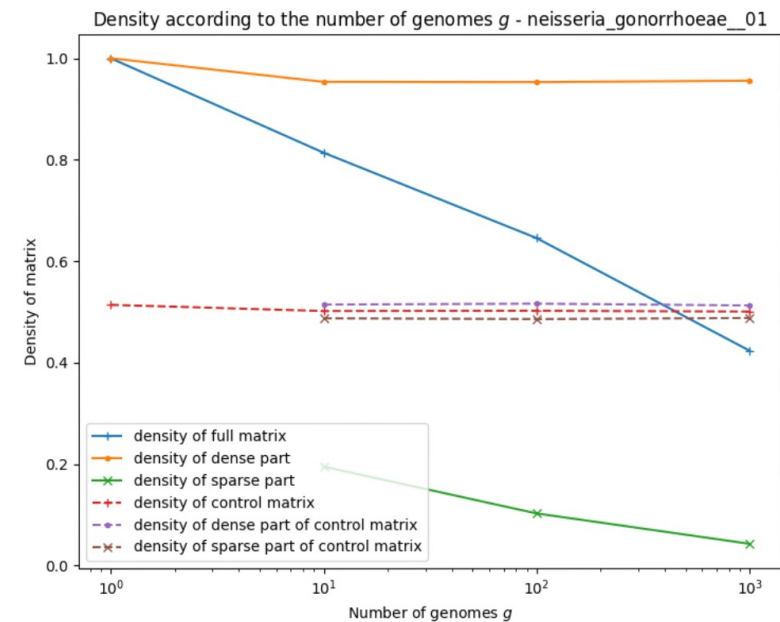
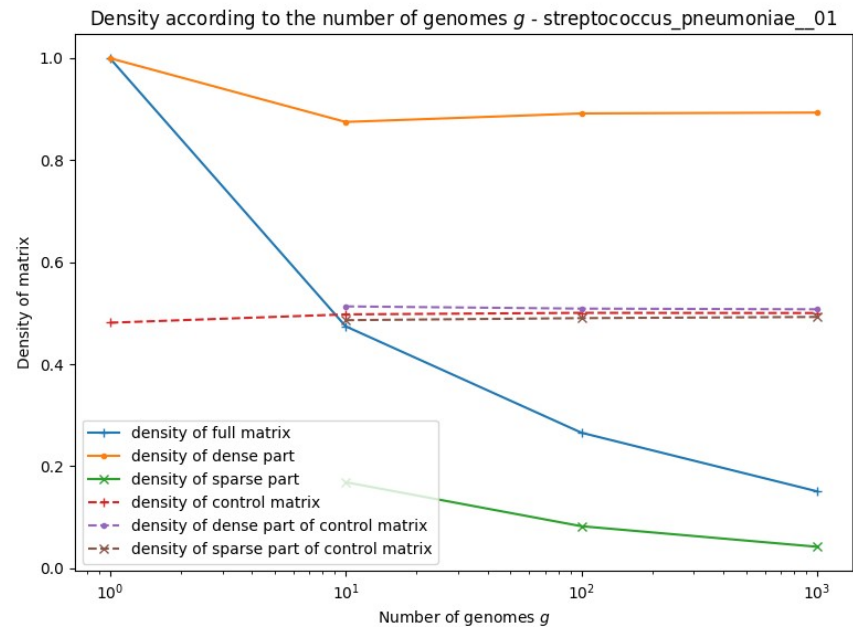
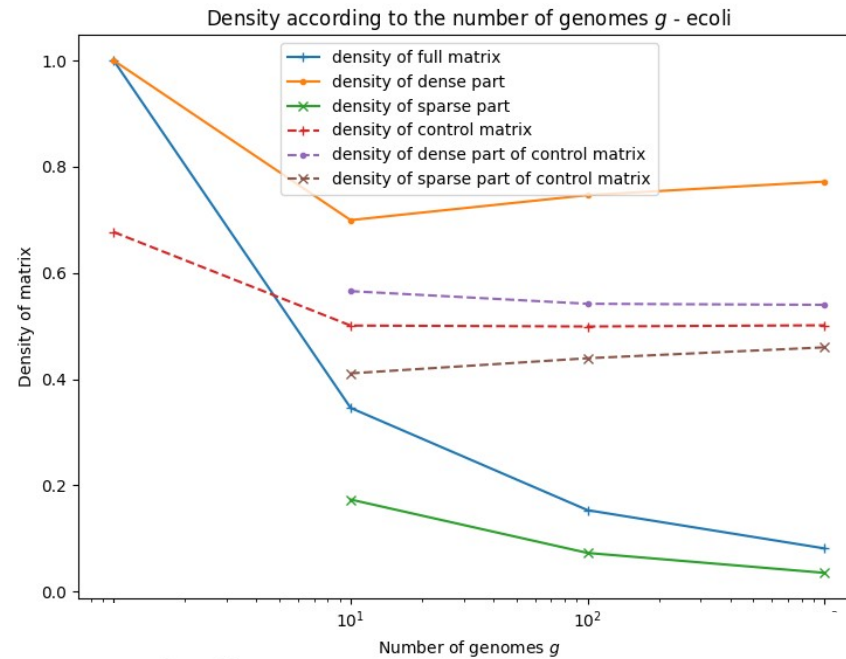
Density of other species

Slightly lower than ngono

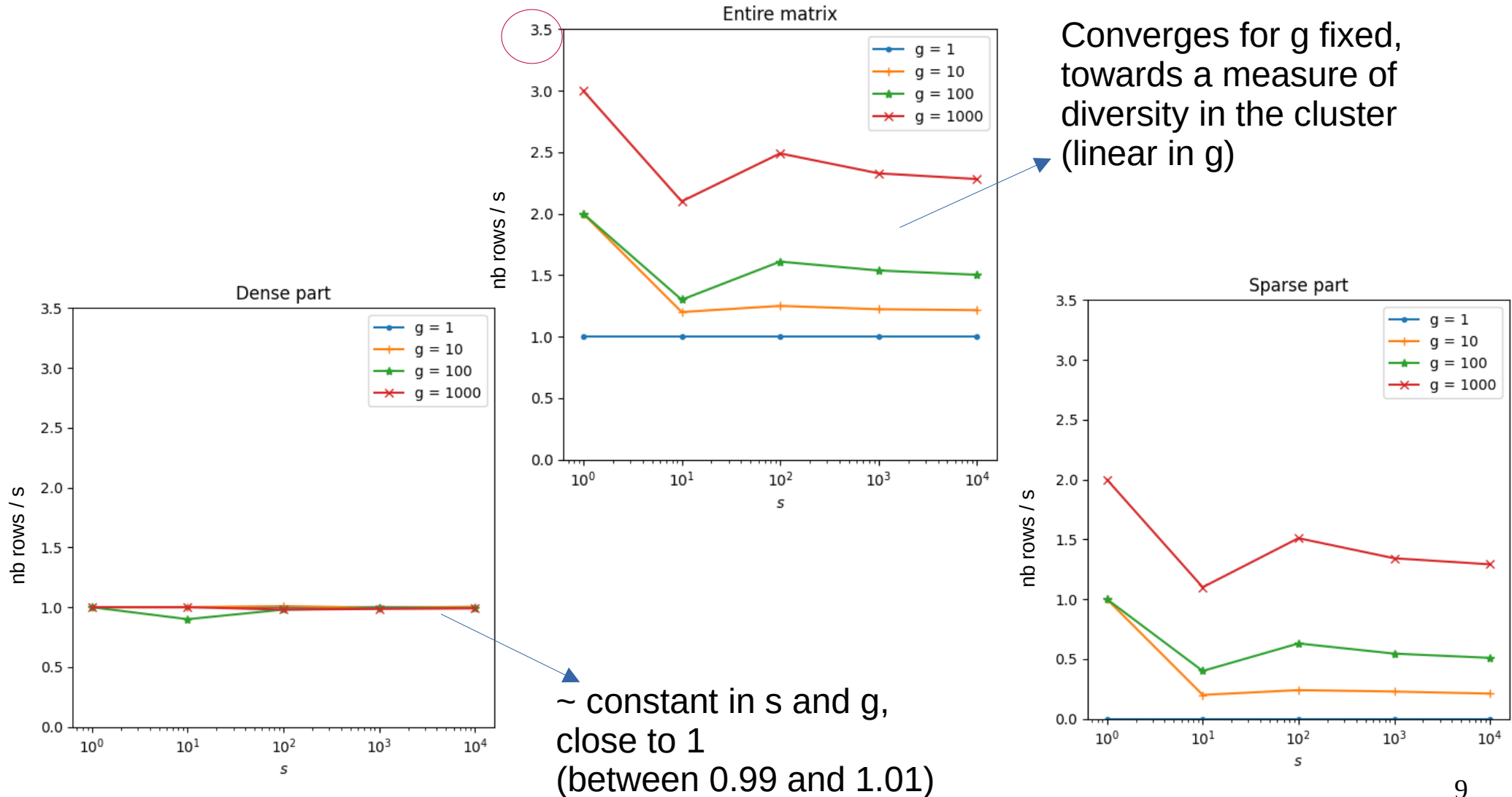


Decreases much more quickly

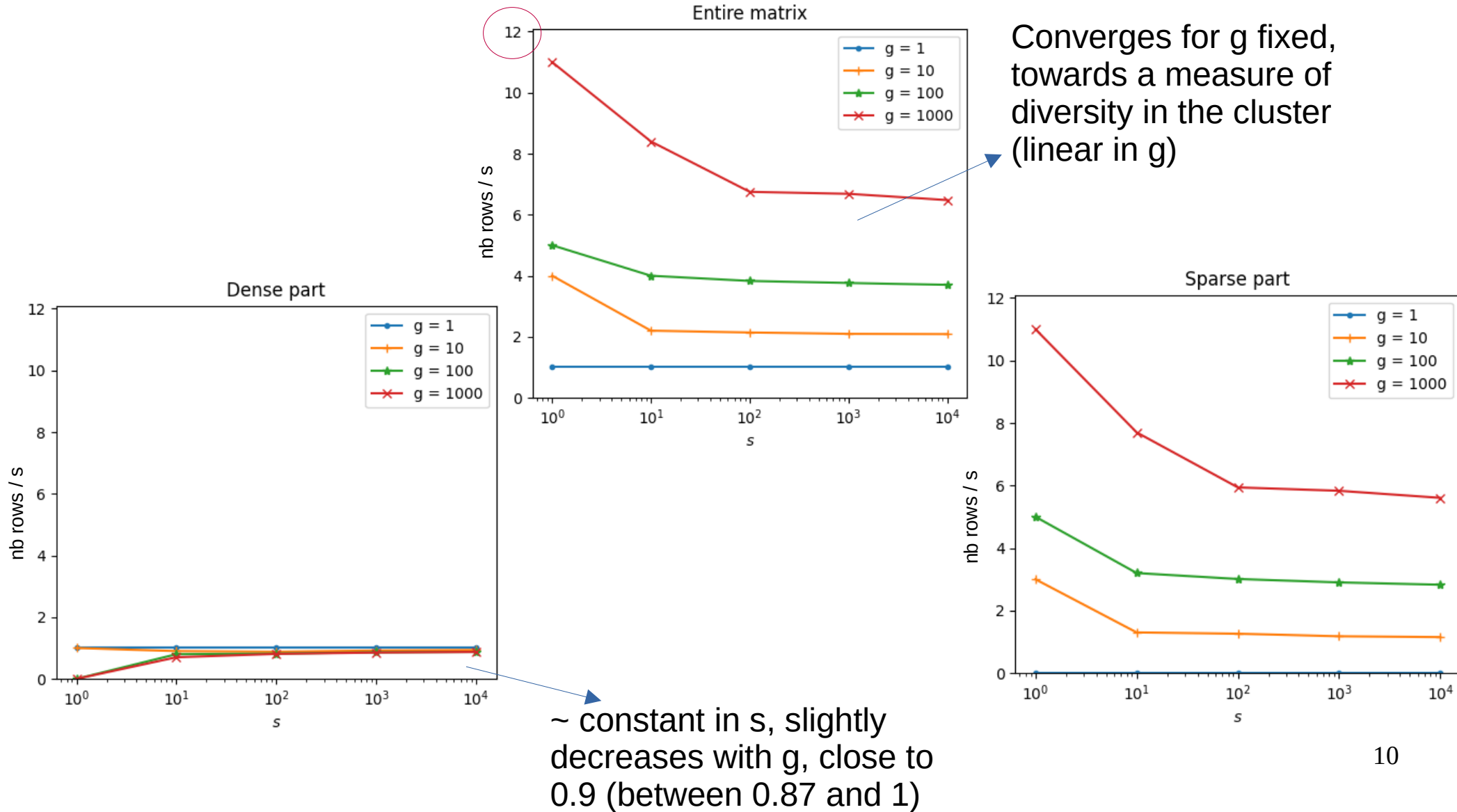
Density of other species



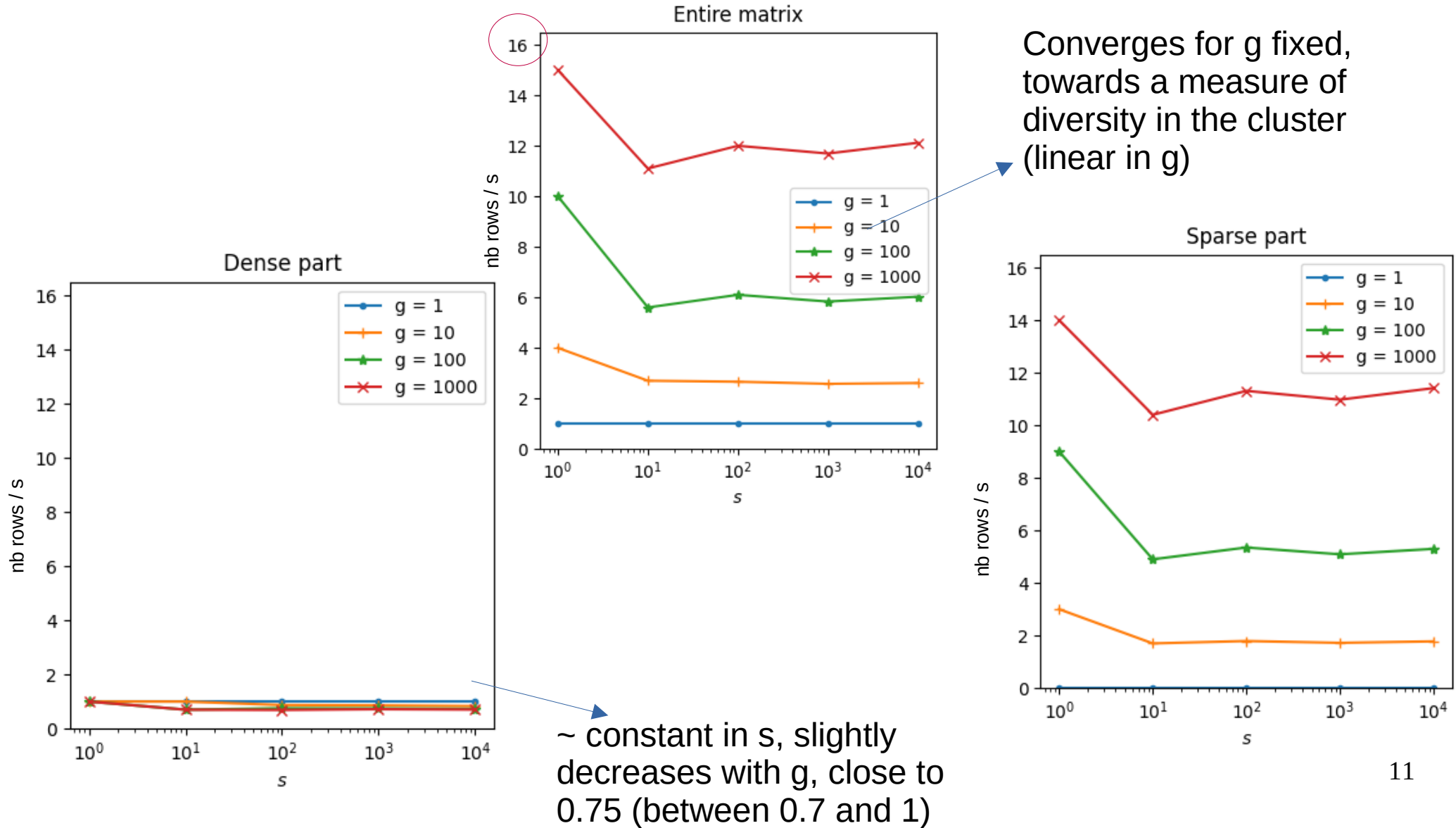
Normalized (by s) number of rows for ngonon



Normalized (by s) number of rows for spneumo



Normalized (by s) number of rows for ecoli



Conclusion

Conclusions :

- The “core” hypothesis looks justified
- This dense core is a big proportion of s
- The number of rows seems linear in $s \cdot g$, even for more diverse clusters

To do :

- Establish a biological framework to see if it could fit with this core idea (ISM)
- Give an estimation of the diversity of the batch of genomes just with the dense part (for different archives and one genome, see to which archive it is closest)